Unifying Theory of Dynamic Aether Mechanics: Magnetic and Electric Fields

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Abstract

Within the Dynamic Aether Mechanics framework, the electric field is identified as the Aether flow vector—the instantaneous direction and magnitude of the medium's displacement—while the magnetic field is the flow angle—the rotational geometry or vorticity of that flow. The flow vector is sourced by coherent, net-zero-flow pressure waves emitted by axial oscillation of the bound core ring of each charged particle, with charge sign determined by whether the ring co-rotates (electron) or counter- rotates (positron) with its surrounding vortex. This paper develops the electromagnetic mechanism on this basis, derives the inverse-cube falloff of magnetic dipole fields, explains ferromagnetism through Coulomb energy and Pauli exclusion, and accounts for the Meissner effect. Electromagnetic unification follows naturally: a stationary charge produces a spherically symmetric wave field (**E** only), but motion Doppler-tilts the wavefronts so that angular structure emerges (**B** appears), with the conversion factor v^2/c^2 recovered from two first-order Doppler factors. The identity $\nabla \cdot \mathbf{B} \equiv 0$ becomes a mathematical tautology $\nabla \cdot (\nabla \times \mathbf{v}) \equiv 0$, explaining why magnetic monopoles cannot exist.

This paper is part of a series presenting the Unifying Theory of Dynamic Aether Mechanics. For the foundational concepts of Aether binding, the three independent properties (density, pressure, temperature), and the unification of force constants, see Paper 1: General Overview [4].

1. Magnetism

Magnetism in this theory arises from the *angular structure* of Aether flow set in motion by electron spin. Each electron acts as a vortex, spinning the surrounding Aether into a circulatory pattern. While the electric field is the Aether flow vector—the instantaneous direction and magnitude of the medium's displacement in the coherent axial-oscillation wave field of the bound core ring (Section 2)—the magnetic field is the flow angle: the rotational geometry of that velocity field. When two or more electron vortices interact, the angular structure of their flows creates pressure differentials that produce the forces we observe as magnetism. These are Bernoulli (dynamic) pressure differentials: where opposing flows meet, kinetic energy converts to static pressure; where flows cooperate, static pressure converts to kinetic energy. This mechanism operates in the incompressible-flow limit—no density change is involved—making Bernoulli dynamics the natural description of all electromagnetic forces in a nearly incompressible superfluid. In the two-fluid model (Paper 6 [5]), electromagnetic fields are properties of the *superfluid* component: the electron vortex is a quantized circulation in the condensate, and the electric and magnetic fields are its velocity and angular flow structure. The normal component is electromagnetically nearly decoupled in the sense of energy exchange: direct absorption and scattering of photons by normal-fluid quasiparticles is suppressed by $P_{\text{th}}/K_A \sim 10^{-38}$ (where $P_{\text{th}} \approx \rho_\Lambda c^2 \sim 6 \times 10^{-10}$ J/m^3 is the thermal pressure in cosmic voids and $K_A \sim 10^{28}$ J/m^3; see Paper 6 [5]). This suppression ensures transparency but does not prevent the normal fraction from modifying the background propagation speed through the two-fluid equation of state (Paper 6, Section 5.1).

1.1 The Basic Mechanism

As an electron spins, it causes the Aether around it to spin as well, creating a vortex in the surrounding medium. When two electrons come near each other, the Aether spinning around them begins to interact.

When spinning in the same direction (parallel spins), the Aether between the two electrons is impeded—the flows want to move in opposite directions at the boundary. This creates a region of high pressure between them, pushing them apart.

When spinning in opposite directions (antiparallel spins), the Aether between them flows in the same direction, allowing it to move freely and creating a region of low pressure. The surrounding higher pressure then pushes the electrons together.

The result: alike-spinning electrons repel, and opposite-spinning electrons attract. This effect is greatest at the poles of the electron, where a greater surface area of spin interacts at higher velocities, naturally producing the strongest forces along the spin axis.

Note that magnetism arises from the angular structure of interacting electron flows—the pressure differentials created at the boundary between two spinning regions of Aether. This is distinct from both the electric field (which arises from the coherent axial-oscillation wave field of the bound core ring; Section 2) and the gravitational mechanism (which arises from the vortex concentration and transient binding of Aether). All three effects originate from the electron's interaction with Aether, but they involve geometrically distinct aspects of the same medium: the electric field is the flow vector (the instantaneous direction and magnitude of the medium's wave-field displacement), the magnetic field is the flow angle (how the flow pattern rotates), and gravity is a binding phenomenon (how each vortex concentrates and cycles Aether independently).

1.2 Field Topology: Closed Lines and No Monopoles

In this model, the Aether flow around each electron forms a vortex—Aether drawn inward along the spin axis, circulated outward along the equatorial plane (or vice versa), creating a continuous loop. The "magnetic field" is really a map of the angular structure of this flow pattern—the vorticity of the Aether velocity field.

This immediately explains why magnetic field lines always form closed loops. The vortex is a circulatory flow: Aether exits one pole, curves around through the surrounding space, and returns at the opposite pole. There is no source or sink—only circulation. A magnetic monopole would require a point from which Aether flow radiates outward in all directions without returning, which is not possible for a vortex. The topology of rotational flow inherently forbids monopoles.

In the flow-vector/flow-angle framework, this result is even stronger. The magnetic field is the vorticity of the Aether velocity field: $\mathbf{B} \propto \nabla \times \mathbf{v}$. The divergence of any curl vanishes identically—$\nabla \cdot (\nabla \times \mathbf{v}) \equiv 0$—so $\nabla \cdot \mathbf{B} = 0$ is not a physical law requiring explanation but a mathematical identity. Magnetic monopoles are forbidden by the geometry of the medium, not by an empirical regularity.

This is consistent with the experimental fact that no magnetic monopole has ever been observed, despite extensive searches. Every magnet, from a single electron to a superconducting coil, has two poles connected by closed field lines.

1.3 Field Falloff: The Inverse Cube Law

The magnetic field of a single electron (a magnetic dipole) falls off as the inverse cube of distance ($B \propto 1/r^3$), in contrast to gravity's inverse square law ($F \propto 1/r^2$). This difference emerges naturally from the vortex model.

Gravity is produced by the monopole effect of transient binding: each atom produces a spherically symmetric low-pressure pulse that dissipates over the surface area of an expanding sphere ($4\pi r^2$), giving $1/r^2$ falloff. The gravitational source is isotropic—it has no preferred direction.

Magnetism is produced by a dipole—a vortex with two poles and opposing flow directions. The flow pattern has both radial and angular structure. At any given distance, the flows from the two poles partially cancel (the inward flow at one pole is offset by the outward flow at the other), and this cancellation becomes more complete with distance. The net effect is an additional factor of $1/r$ beyond the geometric $1/r^2$, producing the observed $1/r^3$ falloff.

For macroscopic magnets—where the Aether flows from enormous numbers of aligned electron vortices combine—the field is more complex at close range but still follows dipole falloff at distances large compared to the magnet's size. This is consistent with observation.

1.4 Ampère's Law: Moving Charges and Magnetic Fields

A moving electric charge produces a magnetic field, as described by Ampère's law. In this theory, the mechanism connects directly to the electric field model described below: each charge emits a coherent axial-oscillation wave field (the electric field), and when the charge moves, the emitted wavefronts are Doppler-tilted—blue-shifted ahead, red-shifted behind—producing a net azimuthal phase gradient around the axis of motion. This Doppler-induced anisotropy gives the wave field a nonzero time-averaged vorticity along the direction of travel—creating the magnetic field observed around current-carrying conductors.

This Doppler-tilted wave pattern produces a circular rotational structure centered on the direction of travel—exactly the circular magnetic field observed around a current-carrying wire. The faster the charge moves, the greater the Doppler anisotropy, and the stronger the resulting magnetic field. For a wire carrying current (many moving electrons), the individual circulatory effects sum, producing a net circulatory Aether flow around the wire.

This is consistent with the right-hand rule: the orientation of the angular structure is determined by the direction of motion relative to the flow vector. The superposition of these flows produces the characteristic field pattern observed around current-carrying conductors.

1.5 Additivity and Material Permeation

Magnetism is additive. Pressure is also additive. The net sum of aligned electron spin vortices determines the total Aether flow pattern and, consequently, the total magnetic field strength. When more spins align, their vortex flows reinforce, producing proportionally stronger pressure differentials.

Magnetic fields permeate non-magnetic materials. Pressure does not require actual flowing Aether to transmit force. A non-magnetic material would disrupt the directional flow of Aether (with randomly oriented electrons each redirecting the flow), but the net sum of pressure would still propagate, creating magnetic forces beyond the material. The Aether between and around the material's atoms still transmits the pressure wave from the external vortex, just as sound pressure passes through a wall despite being scattered by the wall's structure.

1.6 Ferromagnetism: Domain Alignment and Exchange Interaction

In ferromagnetic materials (iron, nickel, cobalt), the electron spins of neighboring atoms spontaneously align within microscopic regions called magnetic domains. In the Aether model, this alignment is energetically favorable because antiparallel-spinning neighbors create low-pressure channels between them, drawing the atoms into cooperative flow patterns. Once a cluster of spins aligns, the collective vortex flow reinforces further alignment of adjacent atoms, allowing the domain to grow.

When an external magnetic field is applied, these domains reorient to align with the external Aether flow, and the material becomes strongly magnetic. When the field is removed, the domain structure can persist—producing a permanent magnet, where the frozen alignment of electron vortices maintains a self-sustaining Aether flow pattern.

The strength of ferromagnetic exchange coupling is naturally explained by the combination of two mechanisms developed elsewhere in this theory. The exchange interaction that drives ferromagnetic alignment is not an amplified magnetic dipole effect—it is fundamentally a *Coulomb energy difference* mediated by the Pauli exclusion principle.

When neighboring atoms' valence electrons have parallel spins, the topological vortex exclusion principle (Section **??**) forces them into antisymmetric spatial configurations: same-spin electron vortices cannot overlap in the same spatial mode, so they remain farther apart on average. This greater separation reduces their mutual Coulomb repulsion (transmitted through the coherent axial-oscillation wave field, Section 2), lowering the total energy. For antiparallel spins, no such topological constraint exists—the electrons can approach more closely, increasing Coulomb repulsion.

The energy scale of this effect is set by the Coulomb interaction (~1 eV per atom pair), not by the magnetic dipole coupling ($\sim 10^{-4}$ eV). This explains the observed factor of ~10,000 between exchange and dipole energies without requiring any amplification mechanism—the two effects operate through entirely different Aether processes (coherent axial-oscillation wave pressure versus circulatory vortex coupling). The crystal-structure dependence of ferromagnetism follows because the exchange energy depends sensitively on orbital overlap geometry: only certain lattice spacings and electronic configurations produce a net energy gain from parallel alignment.

1.7 Curie Temperature: Thermal Disruption of Alignment

A ferromagnetic material loses its magnetism when heated above a characteristic temperature (the Curie temperature). In this theory, heat is the thermal energy of Aether—the random kinetic motion of Aether particles. At low temperatures, this random motion is weak, and the organized vortex flows of aligned electron spins can maintain coherent flow patterns (magnetic domains).

As temperature rises, the thermal energy of the surrounding Aether increases. The random thermal motions of the Aether begin to disrupt the organized flow patterns, buffeting the electron vortices and breaking their alignment. At the Curie temperature, the thermal disruption is strong enough to overwhelm the cooperative flow coupling that maintains domain structure. Above this temperature, the material becomes paramagnetic—the spins can still individually respond to an external field, but they can no longer maintain spontaneous alignment against the thermal noise.

This is directly analogous to how turbulence disrupts laminar flow in classical fluids. An organized flow pattern (laminar flow / magnetic alignment) can only persist when the driving force (vortex coupling) exceeds the disrupting force (thermal turbulence). The Curie temperature marks the point of transition from ordered to disordered Aether flow.

The sharpness of this transition in many materials—where magnetism drops abruptly rather than gradually—is consistent with a cooperative phenomenon: as some spins lose alignment, the collective vortex flow weakens, making it easier for thermal motion to disrupt remaining aligned spins. The transition cascades, producing the sharp onset observed experimentally.

1.8 Paramagnetism

In paramagnetic materials, individual electron spins can partially align with an applied magnetic field, but they cannot maintain spontaneous alignment in the absence of the field. In the Aether model, this occurs because the thermal energy of the Aether is strong enough to randomize the spin orientations once the organizing external flow is removed, but not strong enough to prevent temporary alignment while the field is present.

The applied magnetic field represents an organized Aether flow that biases the orientation of electron vortices. Each vortex tends to align its axis with the external flow (because alignment reduces pressure between the vortex and the surrounding flow), but thermal fluctuations continuously knock it out of alignment. The result is partial, statistical alignment that increases with field strength and decreases with temperature—exactly as observed.

The inverse temperature dependence of paramagnetic susceptibility (Curie's law [2]) follows directly: higher temperature means more vigorous thermal disruption, less net alignment per unit field strength, and weaker paramagnetic response.

1.9 Diamagnetism

Diamagnetism is the weak repulsion that all materials exhibit in a magnetic field. It is the weakest form of magnetic response and is typically masked by stronger paramagnetic or ferromagnetic effects when present.

In the Aether model, diamagnetism arises from the response of the Aether flow within atoms when an external magnetic field (organized vorticity envelope) is applied. The external vorticity phase-modulates the axial oscillation of the atom's paired electrons asymmetrically. Because electrons exist in paired states with opposing spins (in filled orbitals), their coupled wave fields already produce balanced, canceling emissions. When an external vorticity is imposed, it advances the axial-oscillation phase of the ring aligned with it and retards the phase of the opposing ring. The retarded ring loses less coherent pressure to the coupling, while the advanced ring gains more—the net result is a small back-pressure that opposes the applied field.

This explains why diamagnetism is universal (all atoms have electron pairs), weak (it is a second-order perturbation of already-balanced flows), and always opposing (the back-pressure always resists the external field regardless of its direction).

1.10 The Meissner Effect: Superconductivity and Complete Flux Expulsion

In superconductors, magnetic fields are completely expelled from the interior of the material (the Meissner effect). In the Aether model, this has a clear interpretation.

At superconducting temperatures, electrons form Cooper pairs—bound pairs of electrons with opposite spin. In Aether terms, each Cooper pair consists of two electron vortices spinning in opposite directions, whose coherent wave-field emissions precisely cancel. The Aether within and around the pair carries, on average, zero net coherent wave field and zero net vorticity.

When these Cooper pairs condense into a coherent superconducting state, the entire interior of the material becomes a region of zero net coherent wave field. An external magnetic field (organized vorticity envelope) cannot penetrate this region because there are no individual, unbalanced wave sources for the external vorticity to couple to. The external field is forced around the material instead, producing the observed expulsion of magnetic flux.

The characteristic penetration depth of the magnetic field into a superconductor (the London penetration depth) corresponds, in this model, to the distance over which the external vorticity envelope is attenuated by interaction with the Cooper pair condensate at the surface. The field doesn't stop instantaneously at the surface—it decays exponentially over a characteristic length scale, consistent with observation.

1.11 The Electron Magnetic Moment: Antiparallel Orientation and the g-Factor

Experiments reveal that the electron's magnetic moment is antiparallel to its spin angular momentum—it points in the opposite direction. This is a quantitative constraint that the Aether model must accommodate.

In the vortex model, the spin angular momentum of the electron corresponds to the rotation of the electron itself (or its associated Aether structure). The magnetic moment, however, is determined by the circulatory Aether flow pattern the spin produces in the surrounding medium. These need not point in the same direction. If the electron's spin drives Aether inward along its spin axis and outward along the equatorial plane, the resulting circulation pattern (which determines the magnetic moment direction) would be opposite to the spin direction—consistent with the observed antiparallel relationship.

The magnitude of the electron's magnetic moment is characterized by the g-factor. The Dirac equation predicts $g = 2$ for a point particle, but the experimentally measured value is $g = 2.00231930436118$ (known to approximately thirteen significant digits [3]). The deviation from 2—the anomalous magnetic moment—arises from virtual particle interactions (transient binding events) that slightly modify the effective vortex flow pattern around the electron.

This connects directly to the transient binding mechanism central to this theory. The anomalous magnetic moment is calculated in quantum electrodynamics by summing the contributions of virtual particle loops. Each virtual pair that forms near the electron momentarily creates its own small vortex, slightly altering the net Aether flow pattern. The cumulative effect of all transient binding events is a slight enhancement of the magnetic moment beyond the bare vortex value.

The extraordinary precision with which this correction is both calculated and measured (agreement to approximately thirteen significant digits) represents both a powerful connection and a demanding constraint: any quantitative formalization of the Aether vortex model must reproduce this correction. The fact that it arises from transient binding—the same mechanism this theory proposes for gravity—suggests a deep connection between magnetic and gravitational phenomena, unified through the vortex dynamics of the electron.

2. The Electric Field: A Phase-Locked Acoustic Force

The electric field—the force between charged particles described by the Coulomb interaction—has a natural interpretation in the Aether framework as a *phase-locked acoustic force* between two coherent monopole sources in the medium. The electric field is identified with the *Aether flow vector*: the instantaneous direction and magnitude of the medium's velocity at each point. What produces this flow vector is a coherent *pressure-wave field* radiated by the charged particle's internal structure—a net-zero-flow oscillation whose time-averaged mass flux is exactly zero.

The force between two such radiators is the *secondary Bjerknes force* [9]: the standard fluid-dynamic interaction between two phase-locked acoustic monopoles, which reproduces Coulomb's $1/r^2$ falloff and the attraction/repulsion sign rule directly from classical acoustics. The present paper develops this mechanism *generically*, using only the requirements that each source be a coherent monopole radiator of a well-defined frequency, that the source carry a discrete topological sign invariant, and that two sources phase-lock via driven-oscillator inversion. All of the Coulomb-force *structure* ($1/r^2$ falloff, signs, near-field reactive character) follows from these generic requirements alone and is independent of the specific topology of the source.

The numerical value of Coulomb's constant k depends on the source's specific geometry and oscillation amplitude and therefore requires a concrete topological realization. Paper 7(a) [7] supplies one such realization—the bound core ring of a charged lepton, a quantized vortex ring with a toroidal ring of bound Aether at its core—and derives k quantitatively from ring parameters alone, reproducing the measured value within a few percent with no free parameters. Other topologies that satisfy the four generic requirements would reproduce the same force structure but potentially different numerical coupling; Paper 7(a)'s ring is the realization compatible with the electron's self-consistent mass and form-factor constraints.

Where the electric field is the flow vector (the instantaneous displacement of the wave field), the magnetic field is the *flow angle*: the rotational geometry of the Aether velocity field (how the flow pattern is turning). A stationary charge produces a spherically symmetric standing-wave pattern with zero time-averaged vorticity: **E** only, $\mathbf{B} = 0$. When that charge moves, Doppler-anisotropy of the emitted wavefronts imparts a net azimuthal phase gradient around the axis of motion—angular structure appears, and the magnetic field follows. The magnetic field is not a separate entity but the rotational character of the same coherent wave field that constitutes the electric field.

2.1 Required Properties of the Source

For the phase-locked acoustic force to reproduce the Coulomb interaction, the charge-bearing source inside each particle must satisfy four requirements:

1. **Coherent monopole radiator.** The source must emit coherent spherically-symmetric pressure waves into the Aether, so that the secondary Bjerknes formula applies and the resulting force inherits a $1/r^2$ spatial structure from energy conservation across spherical shells.

2. **Discrete topological sign invariant.** The source must carry a two-valued topological invariant (e.g. orientation, handedness, or a relative winding sign) that distinguishes two charge species, so that phase-locking produces either in-phase or anti-phase coupling depending on whether the two charges share the same invariant.

3. **Characteristic frequency fixed by internal geometry.** The source must oscillate at a well-defined frequency ω set by an internal length scale ℓ via the wave relation $\omega \sim c/\ell$. For any source built from the quantum structure of a charged lepton, dimensional considerations fix ℓ at the reduced Compton length $r_{\text{loop}} = \hbar/(m_e c)$, giving

$$\omega = \frac{c}{r_{\text{loop}}} = \frac{m_e c^2}{\hbar}, \tag{1}$$

   the Compton frequency of the electron—the same frequency that appears in zitterbewegung, reflecting the common geometric origin of both phenomena in the minimum-uncertainty structure of the charge-bearing state.

4. **Driven-oscillator phase inversion at the source boundary.** When an incoming pressure wave drives the source's internal oscillation mode, the re-radiated outgoing wave must be 180° inverted relative to the driver, so that two same-species charges phase-lock anti-symmetrically (Section 2.4).

The bound core ring of Paper 7(a) [7]—a quantized vortex ring with circulation $\kappa_e = h/(2m_A)$ whose core contains a permanent toroidal ring of bound Aether produced by the Dynamic Casimir

Effect at the vortex boundary—satisfies all four requirements. The ring radiates coherently via axial oscillation of its center of mass against the Bernoulli restoring pressure of the core; it carries a topological rotation-coupling sign (co-rotating vs. counter-rotating with the surrounding vortex, Section 2.2); its loop radius is fixed at $r_{\text{loop}} = \hbar/(m_e c)$ by Paper 7's circulation-quantization argument; and driven-oscillator inversion follows from standard acoustic boundary conditions at the ring surface.

The equilibrium axial-oscillation amplitude—the single quantitative ingredient beyond geometry—is derived in Paper 7(a) [7] from a healing-length argument, and that derivation reproduces the measured value of Coulomb's constant k to within a few percent from Aether parameters alone. We therefore treat the concrete numerical value of k as established by Paper 7(a) and develop the remainder of the present section in its generic form, applicable to any topology satisfying the four requirements above.

2.2 Topological Charge: Co-Rotation vs. Counter-Rotation

Requirement 2 of Section 2.1—a discrete binary topological invariant distinguishing the two charge species—is realized in the bound core ring topology of Paper 7(a) as the *relative rotation coupling* between the ring and its surrounding vortex. The ring and vortex may rotate in the same sense (co-rotating) or in opposite senses (counter-rotating). This relative rotation coupling is a discrete, topological property of the bound system—there is no continuous deformation between the two configurations—and it defines the sign of charge:

- **Electron** (co-rotating ring/vortex): negative charge, $q = -e$.
- **Positron** (counter-rotating ring/vortex): positive charge, $q = +e$.

Because the coupling is purely topological (a relative winding sign), charge has no spatial extent. The spherical symmetry of the electric field at large distances is exact, not the result of angular averaging over a distributed ring source. This resolves the topological-vs-geometric charge question from Paper 7(a) [7] in favor of the topological interpretation: $F(Q^2) = 1$ exactly, consistent with electron-scattering form-factor measurements at all tested momentum transfers.

Unlike the earlier radial-flow picture, the mechanism requires no net source or sink of Aether at the particle—both electrons and positrons are net-zero-flow structures, distinguished only by the relative winding of ring and vortex.

The stability of charge sign follows directly from the topological structure of the two constituents. The bound core ring's winding number is fixed: it cannot change without destroying the particle, because any continuous deformation must preserve the topological invariant of the bound circulation. The surrounding host vortex's flow direction, by contrast, can be reversed—but only through a high-energy event that decouples the core ring from the host vortex (pair production and annihilation are the physical realizations of such decoupling). In all ordinary conditions, both components retain their formation configuration, so the sign of the ring–vortex rotation coupling—and therefore the sign of charge—is stable over the particle's lifetime.

2.3 Piston-Mode Radiation: The Axial-Oscillation–Vortex Interaction

In the bound core ring realization of Paper 7(a) [7], the coherent monopole radiator of Requirement 1 takes a specific form: the ring oscillates along its symmetry axis, pushing and pulling the surrounding Aether in a coherent, spherically symmetric pattern—a *piston-mode radiator* in the acoustic sense, driving monopole pressure waves into the medium at frequency $\omega = m_e c^2/\hbar$.

The character of the emitted wave depends on the rotation coupling of Section 2.2:

- **Electron (co-rotating).** On the half-cycle when the ring moves in the direction that aligns with the surrounding vortex flow, the two motions cooperate; the boundary moves into a lower-impedance region and the compression half of the cycle is weaker than the rarefaction half. The net emission leads with a rarefaction—a *low-pressure wave*.
- **Positron (counter-rotating).** The axial motion opposes the vortex flow; the boundary enters higher impedance on the compression half, which dominates. The net emission leads with a compression—a *high-pressure wave*.

The sign of the emitted wave's leading pressure phase is therefore the macroscopic signature of charge sign. No steady radial flux of Aether is required; the emission is a coherent oscillation whose time-averaged mass transport is zero, consistent with local Aether conservation.

2.4 Self-Perpetuation and Phase-Locking

The axial oscillation does not need an external driver. The incoming pressure wave from another charge *is* the driver: its pressure cycle forces the receiver's ring along its axis, and the receiver's resulting axial motion re-radiates its own outgoing wave. Because the ring is a driven oscillator at a restoring-force boundary, the outgoing wave is inverted relative to the driving wave—their pressure cycles are 180° out of phase. Two charges of the same species therefore sit in a phase-locked configuration: each emits a wave that is perfectly out of phase with the wave it receives.

This coupled-oscillator regime is what distinguishes the Coulomb interaction from other possible wave couplings in the medium. The two charges are not independent emitters; they form a coherent resonant pair whose force follows from the fixed phase relationship of their mutual wave fields.

2.5 Attraction and Repulsion from Wave Interference

The Coulomb force arises from the time-averaged pressure distribution produced by the superposition of the two charges' coherent wave fields.

- **Like charges repel.** Two electrons emit rarefaction-leading waves that are 180° out of phase with each other (Section 2.4). Between the charges, the two waves destructively interfere: the rarefaction troughs of one cancel the rarefaction troughs of the other, leaving the intermediate region at ambient pressure. Outside this region, each particle sits inside its own coherent low-pressure envelope. The ambient-vs-low-pressure differential pushes the particles apart—repulsion. The same argument with signs reversed applies to two positrons: their compression-leading waves cancel between them, leaving ambient pressure between and each particle inside its own high-pressure envelope; the pressure differential again pushes the particles apart.
- **Opposite charges attract.** A rarefaction-leading wave (electron) and a compression-leading wave (positron) are again 180° out of phase, but the opposite pressure signs of the two waves invert the effective phase, producing *constructive* interference between the charges. The intermediate region carries a strongly modulated pressure oscillation, whose time-averaged effect—owing to the coherent amplitude reinforcement—is a net low-pressure channel between the charges. The surrounding higher pressure pushes them together—attraction.

This is the same Bernoulli-style mechanism that underlies magnetism (parallel spins repel through high pressure between, antiparallel spins attract through low pressure between), but operating on the coherent wave field of ring axial oscillation rather than the rotational flow pattern of the spin vortex.

The mapping onto established physics is direct: this is the *secondary Bjerknes force* between two phase-locked acoustic radiators, first described by V. F. K. Bjerknes in the early twentieth century [9]. Two phase-locked pulsating sources in a fluid medium are known to attract when oscillating in phase and repel when oscillating out of phase, with a $1/r^2$ falloff—exactly the sign structure and distance dependence of the Coulomb force. The new mechanism therefore inherits its attraction/repulsion behavior directly from a classical, experimentally verified acoustic phenomenon, rather than requiring any novel assumption about charge interaction. The Bjerknes force is a *reactive, near-field* coupling—it arises from mutual radiation impedance between the sources, not from far-field radiation transport—which fits naturally with the near-field regime of Coulomb interactions established in Section 2.7: classical Coulomb forces carry no net radiation to infinity because they are reactive wave couplings, not radiative ones.

2.6 The Electron's Three Aether Effects

With the electric field defined as the flow vector (the instantaneous displacement of the axial-oscillation wave field) and the magnetic field as the flow angle, each electron produces three distinct

Aether phenomena, each from a different aspect of its interaction with the medium:

1. **Axial-oscillation wave field** (coherent monopole radiation from the bound core ring)—produces the electric field. Determines charge sign (via ring/vortex rotation coupling) and Coulomb interactions. Falls off as $1/r^2$ (spherical spreading of wave intensity).
2. **Flow angle** (circulatory, from spin)—produces the intrinsic magnetic moment. Determines magnetic dipole interactions with other spinning particles. Falls off as $1/r^3$ (dipole cancellation).
3. **Vortex-boundary pair creation and transient binding**—produces gravity. The vortex core acts as a rapidly moving boundary (the DCE mechanism below the permanence threshold), producing the transient bind–unbind pressure wave. Falls off as $1/r^2$ (spherical spreading of the monopole pressure wave), but is enormously weaker than the electric force for reasons developed in Section 2.8.

These three effects coexist because they involve geometrically distinct aspects of the Aether velocity field: the coherent axial-oscillation wave (magnitude and direction of the instantaneous displacement), the flow angle (rotational geometry), and the DCE-driven binding cycle. They are not in competition—a single electron sustains all three simultaneously.

2.7 Falloff: $1/r^2$ from Spherical Wave-Intensity Spreading

A coherent spherical pressure-wave radiator has time-averaged intensity $\langle I \rangle \propto 1/r^2$ by conservation of wave energy through spherical shells of area $4\pi r^2$. The corresponding pressure amplitude falls as $p(r) \propto 1/r$, and the force between two phase-locked radiators follows the product of amplitudes weighted by the coherence factor—recovering the $1/r^2$ Coulomb falloff identically to the earlier radial-flow picture, but from wave-intensity spreading rather than flow-vector spreading.

This is the near-field (quasi-static) limit where the charge separation r is well within the wavelength $\lambda = 2\pi c/\omega$. For the Compton-scale oscillation frequency adopted in Section 2.1, λ is of order 10^{-12} m, so all macroscopic Coulomb interactions lie deep in the near-field regime, and Coulomb's law holds across all classical distances.

2.8 Coulomb vs. Gravity: Four Independent Differences

The enormous strength of the Coulomb force compared to gravity (a ratio of order 10^{42} for two electrons) follows from the product of four independent differences between the axial-oscillation wave field (electric) and the bind–unbind residual pressure wave (gravity):

1. **Amplitude.** The Coulomb wave amplitude is set by the full axial displacement A of the bound core ring. The gravity wave amplitude is the residual asymmetry of the transient bind–unbind cycle at the vortex core boundary—a small fraction of the oscillation-scale amplitude.

2. **Frequency.** The Coulomb wave oscillates at the ring's axial-oscillation frequency $\omega = m_e c^2/\hbar$ (Section 2.1). The gravity wave oscillates at the DCE bind–unbind cycling frequency at the vortex boundary—a different characteristic frequency set by the permanence threshold velocity v_{perm} and the healing length ξ (Paper 3 [8], Paper 7(a) [7]).

3. **Asymmetry.** The axial-oscillation emission is a symmetric sinusoid whose positive and negative half-cycles carry full, coherent, phase-locked force. The gravitational emission is asymmetric—the residual after nearly complete bind–unbind cancellation—and carries only the small unrecycled component.

4. **One-way vs. two-way coupling.** Two charges *mutually* drive each other's axial oscillation: each ring's oscillation is the receiver's response to the incoming wave, creating the phase-locked coupled-oscillator regime of Section 2.4 that underlies the Coulomb force. Gravity, by contrast, may be below the receiver's response threshold—the bind–unbind residual wave amplitude may be too small to drive any resonant back-reaction in a receiver vortex. The interaction is effectively one-way (asymmetric-emission only) rather than a mutual coupled-oscillator. This difference in coupling *character*, not just in amplitude, is a significant contributor to the observed force hierarchy.

The 10^{42} ratio is the product of these four factors. A first-principles determination of each factor—amplitude A, frequency ratio, asymmetry fraction ϵ, and response-threshold criterion—is left as an open question (Section 5).

2.9 Electromagnetic Unification: How Movement Connects E and B

The relationship between electric and magnetic fields—the foundation of Maxwell's equations—emerges naturally from the Aether model once the electric field is identified as the flow vector of a coherent axial-oscillation wave and the magnetic field as its angular structure.

When a charge is stationary, its axial-oscillation wave field is spherically symmetric about the ring center. The instantaneous flow vector points radially outward or inward at each phase of the oscillation, but the time-averaged vorticity of this pattern is zero: there is no magnetic field (beyond the intrinsic spin moment of the particle itself).

When a charge moves with velocity $\mathbf{v}$, the emitted wavefronts are Doppler-shifted—blue-shifted ahead of the motion, red-shifted behind. This kinematic anisotropy of the wave field produces a net azimuthal phase gradient around the axis of motion, so that the time-averaged pattern acquires nonzero vorticity along $\mathbf{v}$. The coherent wave field of a moving charge has $\nabla \times \langle \mathbf{E} \rangle \neq 0$—this is the magnetic field produced by a moving charge, expressed as a purely kinematic consequence of the Doppler structure of the radiated wave. The Doppler shift is first order in v/c, and the magnetic interaction between two moving charges involves a Doppler factor from each, recovering the observed $F_{\text{mag}}/F_{\text{elec}} = v_1 v_2 / c^2$ (Section 6).

This mechanism provides the physical basis for several fundamental electromagnetic relationships:

- **Ampère's law:** A current-carrying wire (many co-moving charges) produces a circular magnetic field around it. In the Aether model, each moving charge's Doppler-anisotropic wave field carries a nonzero azimuthal vorticity component, and these individual contributions sum coherently around the wire axis to produce the observed circular field.
- **The right-hand rule:** The orientation of the Doppler-induced vorticity is determined by the direction of motion relative to the wave field—the geometry of wavefront tilting dictates the resulting flow angle.
- **Relativity of E and B fields:** In standard physics, a pure electric field in one reference frame appears as a combination of electric and magnetic fields in another frame moving relative to the first. In the Aether model, a stationary charge has a spherically symmetric wave field ($\mathbf{E}$ only, no time-averaged angular structure). In a boosted frame, the wavefronts are Lorentz-contracted and phase-tilted, so the 4-wavevector of the emitted wave acquires the transverse components that manifest as $\mathbf{B}$. The frame-dependent mixing of flow vector and flow angle that relativity describes as **E**–**B** mixing is therefore a kinematic consequence of the 4-vector transformation of the emitted wave field.
- **Force proportional to velocity:** The magnetic force on a moving charge ($\mathbf{F} = q\mathbf{v} \times \mathbf{B}$) depends on the charge's velocity. In the Aether model, the pressure differential created by the interaction between two wave patterns depends on the relative Doppler-anisotropy of their sources. A stationary charge in a magnetic field experiences no net force because its wave field has no Doppler tilt to couple to the external angular structure. A moving charge's Doppler-anisotropic wave field creates an asymmetric pressure interaction with the external vorticity, producing the Lorentz force.

2.10 Electromagnetic Induction

Faraday's law of electromagnetic induction—a changing magnetic field produces an electric field, and a changing electric field produces a magnetic field—is the basis for generators, transformers, and electromagnetic wave propagation. The Aether model provides a physical mechanism for both directions of induction:

- **Changing B creates E:** A changing magnetic field is a time-varying vorticity envelope in the Aether flow. As this rotational pattern changes, it buffets the axial oscillation of bound rings in its neighborhood and drives net coherent wave-field amplitude—which is the electric field. The changing flow angle displaces Aether in a pattern that produces a net flow vector, inducing an electric field in the surrounding region.
- **Changing E creates B:** A changing electric field is a time-varying coherent wave-field amplitude. Any time-varying wave source is equivalent (via the source-advection mass current) to a moving charge, whose Doppler-anisotropic wavefronts carry azimuthal vorticity—producing a magnetic field. This is Maxwell's displacement current, given a physical interpretation: a changing flow vector develops the same angular character as a moving source.

2.11 Electromagnetic Waves

Under the axial-oscillation picture, the electric field is already a coherent pressure-wave field at every instant, so free-space electromagnetic waves emerge as a direct consequence: a photon is a propagating excitation of the same axial-oscillation wave mode, whose displacement (flow vector) and vorticity (flow angle) components are coupled through the medium's wave equation. The two aspects regenerate each other as the wave propagates—an oscillating flow vector develops azimuthal vorticity (Doppler-anisotropy of the traveling wave), which in turn drives a net transverse flow, which develops vorticity, and so on—at the speed of light.

This is an oscillation in the Aether medium between two aspects of the same coherent wave field—its displacement amplitude/direction and its rotational geometry. The electromagnetic wave does not require a separate mechanism from the other Aether phenomena; it is simply the free-space propagation of the same wave that a stationary charge already carries in bound, near-field form.

The polarization of electromagnetic waves corresponds to the orientation of the flow-vector oscillation. The fact that electromagnetic waves are transverse (oscillating perpendicular to the direction of propagation) is consistent with the Aether model: the flow vector and flow angle disturbances that constitute the wave are perpendicular to each other and to the direction of propagation, just as observed. At the microscopic level, these wave modes are spin-wave excitations of the Aether's spinor condensate (Paper 7 [6]); through spin–orbit coupling in the condensate [10], they produce effective mass currents that manifest as the velocity and vorticity fields described here. The ring's axial oscillation is the bound analog of exactly this spin-wave excitation—a localized, standing-wave form of the same fundamental degree of freedom that propagates as free radiation.

3. Summary of Theoretical Properties

3.1 Properties of Magnetism

- The magnetic field is the angular structure (flow angle) of the Aether velocity field, arising from pressure differentials between interacting electron spin vortices. This is distinct from gravity's binding mechanism. Magnetism is how the rotational geometry of two flows affects the Aether between them; gravity is how each vortex concentrates and cycles Aether independently.
- Parallel spins repel (opposing flows create high pressure between them); antiparallel spins attract (cooperative flows create low pressure between them).
- Field lines form closed loops as a topological consequence of vortex circulation. Magnetic monopoles are forbidden by a mathematical identity: $\nabla \cdot (\nabla \times \mathbf{v}) \equiv 0$.
- Dipole field falls off as $1/r^3$ (partial cancellation of opposing poles at distance), compared to gravity's monopole $1/r^2$ falloff.
- Moving charges produce magnetic fields because Doppler-anisotropy of the emitted axial-oscillation wave field gives a time-averaged vorticity along the axis of motion—kinematic tilting of wavefronts converts flow vector into flow angle (Ampère's law).
- Ferromagnetic alignment occurs when parallel-spin vortex topology forces electrons into antisymmetric spatial configurations, reducing their Coulomb repulsion. The exchange energy ($\sim$1 eV) is primarily a Coulomb effect mediated by Pauli exclusion (antisymmetric spatial wavefunctions reduce electron–electron repulsion), not an amplified magnetic dipole interaction.
- Curie temperature marks the transition where thermal Aether turbulence overwhelms vortex coupling, destroying domain alignment—analogous to the laminar–turbulent transition in classical fluids.
- Paramagnetic response reflects partial, statistical alignment of individual vortices against thermal disruption; diamagnetic response reflects the back-pressure from perturbation of balanced electron-pair flows.
- Superconducting flux expulsion (Meissner effect) occurs when Cooper pairs cancel all net Aether flow, creating a zero-flow interior that excludes external magnetic fields.
- The electron's magnetic moment is antiparallel to its spin, consistent with the vortex model's prediction that the circulation pattern (moment) opposes the rotation direction (spin angular momentum).

- The anomalous magnetic moment (g-factor deviation from 2) arises from transient binding events modifying the effective vortex flow pattern—connecting magnetism quantitatively to the same virtual particle mechanism that produces gravity.

3.2 Properties of Electromagnetism

- The electric field is the Aether flow vector: the instantaneous direction and magnitude of the medium's displacement in the coherent axial-oscillation wave field of the bound core ring. Charge sign is determined by the rotation coupling between ring and surrounding vortex (co-rotation = electron, counter-rotation = positron).

- The magnetic field is the flow angle: the rotational geometry (vorticity) of the Aether velocity field. $\nabla \cdot \mathbf{B} = 0$ is a mathematical identity.

- The Coulomb force arises from pressure differentials between the coherent wave fields of two charges: same-sign wave fields (both out of phase with each other by 180°) destructively interfere between the charges and produce repulsion; opposite-sign wave fields constructively interfere (via sign inversion) and produce attraction.

- The electric force falls off as $1/r^2$ (spherical spreading of wave intensity from a coherent radiator), identical in geometric origin to gravitational falloff.

- The enormous strength of the electromagnetic force relative to gravity reflects four independent differences between the axial-oscillation wave (Coulomb) and the bind–unbind residual wave (gravity): amplitude, frequency, emission asymmetry, and whether the receiver can resonantly respond (two-way vs. one-way coupling).

- Each electron produces three coexisting Aether effects: coherent axial-oscillation wave field (electric field), flow angle (magnetic moment), and transient binding (gravity).

- Magnetism from current arises when the flow vector (electric field) of a moving charge acquires Doppler-anisotropy, giving nonzero time-averaged vorticity along the axis of motion—unifying **E** and **B** as two aspects of the same coherent wave field.

- Electromagnetic induction occurs because changing flow angle (changing **B**) drives a net flow vector (creates **E**), and changing flow vector (changing **E**) develops angular structure (creates **B**).

- Electromagnetic waves are free-space propagations of the same coherent axial-oscillation wave mode that a stationary charge carries in bound, near-field form, with displacement and vorticity components regenerating each other at the speed of light.

3.3 Properties of Force Constants

- Only two independent electromagnetic constants exist at the Aether level: the speed of light c (wave propagation speed) and the Coulomb constant k (coherent axial-oscillation wave-pressure response).
- The magnetic permeability μ_0 is derived: $\mu_0 = 4\pi k/c^2$. It relates the flow-vector coupling to the flow-angle coupling via the medium's wave speed.
- The magnetic force between moving charges equals the electric force scaled by $v_1 v_2/c^2$, where v_1 and v_2 are the charge velocities—recovered from two first-order Doppler factors, one per charge.
- The gravitational constant G introduces one additional parameter beyond the Aether properties in k: ϵ, the fractional asymmetry of the transient bind–unbind cycle.
- The ratio k/G decomposes into the product of four independent factors—wave-amplitude ratio, frequency ratio, emission-asymmetry fraction, and receiver-response threshold—and should be derivable from Aether properties (Section 2.8).
- The factor v^2/c^2 has a quadruple appearance: time dilation ($\gamma = 1/\sqrt{1 - v^2/c^2}$), the magnetic-to-electric force ratio ($F_{\text{mag}}/F_{\text{elec}} = v_1 v_2/c^2$), relativistic energy ($E = \gamma mc^2$), and gravitomagnetism ($B_g = g \times v/c^2$, confirmed by GP-B). All four arise from the same Aether mechanics: v^2/c^2 is the ratio of kinetic energy density to the medium's elastic capacity.
- The Aether is experimentally confirmed to be perfectly linear (Hookean) in its pressure–density response, confirmed by velocity time dilation (to 10^{-8} precision across velocities from $10^{-6}\,c$ to $0.9994c$) and independently by the Fresnel drag coefficient (exact $1 - 1/n^2$ across materials with density ratios up to nearly 6:1). This compressive linearity governs wave propagation; the electromagnetic forces in this paper arise from Bernoulli pressure in the incompressible-flow limit.
- Gravitational time dilation ($\Delta t/t = GM/(rc^2)$) directly connects G to the medium: the Aether inflow velocity $v_{\text{in}} = \sqrt{2GM/r}$ produces $v_{\text{in}}^2/(2c^2) = GM/(rc^2)$, the time dilation of a static observer in the flowing medium.
- The fine structure constant ($\alpha \approx 1/137$) measures the coupling strength of a single electron's axial-oscillation wave field to the quantum scale of the medium.
- The gravitational coupling constant ($\alpha_G \approx 1.75 \times 10^{-45}$) measures the strength of a single electron's residual binding pressure relative to the same quantum scale.
- The Planck mass ($m_P \approx 2.18 \times 10^{-8}$ kg) represents the mass scale at which gravitational binding pressure becomes comparable to quantum-scale Aether disturbances.

4. Experimental Support and Connections to Established Physics

The following experimentally confirmed phenomena support or align with the aspects of the theory presented in this paper:

1. **The anomalous magnetic moment of the electron.** Virtual particle corrections to the electron's magnetic moment have been confirmed to approximately thirteen significant digits [3], establishing that transient particle–antiparticle formation and annihilation is a real, measurable process occurring continuously around charged particles. In this theory, the anomalous moment arises from transient binding events (the same mechanism that produces gravity) modifying the effective Aether vortex flow around the electron—connecting the magnetic and gravitational mechanisms through a single, precisely measured quantity.

2. **Absence of magnetic monopoles.** Despite extensive experimental searches, no magnetic monopole has ever been observed. This is a direct consequence of the vortex model: circulatory Aether flow inherently produces closed field lines with two poles. A monopole would require a topologically impossible radial-only flow from a vortex.

3. **The Meissner effect.** Superconductors expel magnetic fields from their interiors. In this theory, Cooper pairs (opposite-spin electron pairs) cancel all net Aether flow, creating a zero-flow region that external magnetic fields cannot penetrate. The observed exponential decay of the field at the surface (London penetration depth) is consistent with the gradual attenuation of external Aether flow by the Cooper pair condensate.

4. **The $\mu_0 = 4\pi k/c^2$ relationship.** Maxwell's equations [1] establish that the speed of light equals $1/\sqrt{\epsilon_0 \mu_0}$, linking the electric and magnetic constants through c. In the Aether model, this relationship is not a mathematical coincidence but a physical necessity: magnetism is moving electric field, so the magnetic constant must equal the electric constant divided by the square of the propagation speed.

5. Open Questions and Testable Predictions

The following areas relevant to this paper remain underdeveloped and represent opportunities for further refinement or falsification:

1. **Quantitative reproduction of the g-factor anomaly.** The electron's anomalous magnetic moment ($g = 2.00231930436118$) is known to extraordinary precision and is calculated in QED by summing virtual particle loop corrections. If the Aether vortex model can reproduce this correction from first principles—by calculating how transient binding events modify the

effective vortex flow pattern—it would provide one of the most stringent quantitative tests of the theory.

2. **First-principles derivation of the source amplitude and piston area.** The concrete ring topology of Paper 7(a) [7] supplies the missing amplitude through the healing-length closure $A = \xi$ and obtains Coulomb's constant k within a few percent of observation. Two geometric inputs of that derivation remain to be derived from first principles rather than assumed: the effective piston area S_{eff} of the ring's monopole radiation (currently taken as the disc bounded by the ring, πr_{loop}^2) and the saturation of the amplitude at the healing-length bound. A full GP calculation of the axial flow pattern around an oscillating toroidal vortex, and a rigorous derivation of the equilibrium amplitude from the Bernoulli restoring force, would close both sub-problems. Any alternative topology reproducing $\omega \approx m_e c^2/\hbar$ and amplitude $\sim \xi$ with a similar piston-area scaling would also reproduce k within the same tolerance.

3. **Deriving the fine structure constant from Aether properties.** The fine structure constant ($\alpha \approx 1/137$) measures the coupling strength of a single electron's axial-oscillation wave field to the quantum scale of the medium. If the ring's axial-oscillation amplitude and frequency can be calculated from the vortex geometry, and the quantum scale of Aether disturbances is set by $\hbar$ and c, then $\alpha = ke^2/(\hbar c)$ should be derivable from first principles.

6. Mathematical Summary

This section collects the key equations relevant to the topics covered in this paper. *For the complete mathematical summary, see Paper 1: General Overview [4].*

6.1 Fundamental Aether Parameters

The theory posits four microscopic parameters from which all macroscopic phenomena derive:

Symbol	Name	Constrained range	Primary constraint
ρ_A	Ambient density	10^{11}–10^{12} kg/m^3	Electron vortex energy
K_A	Bulk modulus	10^{28}–10^{29} J/m^3	$K_A = \rho_A c^2$
m_A	Aether quantum mass	500–850 MeV/c^2	Flux tube width / glueball mass
ξ	Healing length	0.17–0.28 fm	Bridge diameter (lattice QCD)

The Coulomb-constant cross-check of Paper 7(a) [7], Section "Tightened Parameter Ranges," narrows these ranges further: $m_A \in [628, 689]$ *MeV/c^2,* $\rho_A \in [8.9 \times 10^{11}, 1.07 \times 10^{12}]$ *kg/m^3,* $\xi \in [0.20, 0.22]$ *fm,* $K_A \in [8.0 \times 10^{28}, 9.6 \times 10^{28}]$ *J/m^3.*

These four are not independent. Two defining relations connect them:

$$c = \sqrt{K_A/\rho_A} \tag{2}$$

$$\xi = \frac{\hbar}{m_A\, c\, \sqrt{2}} \tag{3}$$

leaving two free Aether parameters (e.g., ρ_A and m_A) from which all others follow. (A third parameter, ϵ, enters through gravity; see below.)

6.2 Equation of State

The Aether obeys a logarithmic equation of state, the unique form that produces a density-independent bulk modulus (Section **??**):

$$P = K_A \ln\!\left(\frac{\rho}{\rho_0}\right) + P_0 \tag{4}$$

Key consequence: the speed of sound $c_s = \sqrt{K_A/\rho}$ varies with density. At ambient density, $c_s = c$. Note: in gravitational fields, Bernoulli's equation for free-fall inflow gives $\Delta\rho/\rho_A = 0$ exactly, so c does not vary near gravitating masses; the density dependence is relevant for material media and for the two-fluid cosmological structure. The electromagnetic forces described in this paper arise from Bernoulli (dynamic) pressure—the redistribution of kinetic and static energy in flowing Aether—not from this compressive equation of state. The equation of state governs wave propagation and cosmological structure; electromagnetic mechanisms require only the incompressible-flow limit. This equation of state guarantees that K_A is constant under compression, which is experimentally confirmed by:

- Velocity time dilation measurements spanning $10^{-6}\,c$ to $0.9994\,c$ (precision $\sim 10^{-8}$).
- The Fresnel drag coefficient $1 - 1/n^2$, exact across materials with density ratios up to ~6:1.

6.3 Electromagnetic Equations

Electric field (flow vector). The electric field is the Aether flow vector—the instantaneous direction and magnitude of the medium's displacement in the coherent axial-oscillation wave field of the bound core ring (Section 2). The Coulomb force between charges q_1, q_2 at separation r:

$$F_{\text{elec}} = k\,\frac{q_1 q_2}{r^2} \tag{5}$$

Magnetic force (flow angle). Magnetism arises from the angular structure of the Aether flow—the rotational geometry of the velocity field. The magnetic force between charges moving at velocities

v_1, v_2:

$$F_{\mathrm{mag}} = F_{\mathrm{elec}} \frac{v_1 v_2}{c^2} \tag{6}$$

Permeability. The magnetic permeability relates the coherent axial-oscillation wave-field coupling to the flow-angle coupling via the medium's wave speed, and is a derived quantity, not a fundamental constant:

$$\mu_0 = \frac{4\pi k}{c^2} \tag{7}$$

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